

ECE 3317
FINAL EXAM
FORMULA SHEET

General EM

$$\nabla \times \underline{\mathcal{E}} = -\frac{\partial \underline{\mathcal{B}}}{\partial t}$$

$$\nabla \times \underline{\mathcal{H}} = \underline{\mathcal{J}} + \frac{\partial \underline{\mathcal{D}}}{\partial t}$$

$$\nabla \cdot \underline{\mathcal{B}} = 0$$

$$\nabla \cdot \underline{\mathcal{D}} = \rho_v$$

$$\nabla \times \underline{E} = -j\omega \underline{B}$$

$$\nabla \times \underline{H} = \underline{J} + j\omega \underline{D}$$

$$\nabla \cdot \underline{B} = 0$$

$$\nabla \cdot \underline{D} = \rho_v$$

$$\underline{\mathcal{E}}(x, y, z, t) = \text{Re}(\underline{E}(x, y, z)e^{j\omega t})$$

$$\underline{\mathcal{H}}(x, y, z, t) = \text{Re}(\underline{H}(x, y, z)e^{j\omega t})$$

$$\nabla \cdot \underline{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

$$\nabla \cdot \underline{V} = \frac{1}{\rho} \frac{\partial}{\partial \rho}(\rho V_\rho) + \frac{1}{\rho} \frac{\partial V_\phi}{\partial \phi} + \frac{\partial V_z}{\partial z}$$

$$\nabla \cdot \underline{V} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 V_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta}(V_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial V_\phi}{\partial \phi}$$

$$\nabla \times \underline{V} = \hat{x} \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) + \hat{y} \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) + \hat{z} \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right)$$

$$\nabla \times \underline{V} = \underline{\hat{\rho}} \left(\frac{1}{\rho} \frac{\partial V_z}{\partial \phi} - \frac{\partial V_\phi}{\partial z} \right) + \underline{\hat{\phi}} \left(\frac{\partial V_\rho}{\partial z} - \frac{\partial V_z}{\partial \rho} \right) + \underline{\hat{z}} \frac{1}{\rho} \left(\frac{\partial(\rho V_\phi)}{\partial \rho} - \frac{\partial V_\rho}{\partial \phi} \right)$$

$$\nabla \times \underline{V} = \underline{\hat{r}} \frac{1}{r \sin \theta} \left[\frac{\partial(V_\phi \sin \theta)}{\partial \theta} - \frac{\partial V_\theta}{\partial \phi} \right] + \underline{\hat{\theta}} \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial V_r}{\partial \phi} - \frac{\partial(r V_\phi)}{\partial r} \right] + \underline{\hat{\phi}} \frac{1}{r} \left[\frac{\partial(r V_\theta)}{\partial r} - \frac{\partial V_r}{\partial \theta} \right]$$

$$c \equiv 2.99792458 \times 10^8 \text{ [m/s]}$$

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\epsilon_0 \doteq 8.8541878 \times 10^{-12} \text{ [F/m]}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ [H/m]}$$

$$\underline{D} = \epsilon \underline{E}$$

$$\underline{B} = \mu \underline{H}$$

$$\epsilon = \epsilon_0 \epsilon_r$$

$$\mu = \mu_0 \mu_r$$

$$\oint_S (\underline{\mathcal{E}} \times \underline{\mathcal{H}}) \cdot \hat{n} dS = - \int_V \sigma |\underline{\mathcal{E}}|^2 dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \mu |\underline{\mathcal{H}}|^2 \right) dV - \int_V \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon |\underline{\mathcal{E}}|^2 \right) dV$$

$$\underline{\mathcal{J}} = \underline{\mathcal{E}} \times \underline{\mathcal{H}}$$

$$\mathcal{P}(t) = \int_S \underline{\mathcal{J}} \cdot \hat{n} dS$$

$$\langle \mathcal{P}(t) \rangle = \text{Average power flow (watts)}$$

$$\underline{S} \equiv \frac{1}{2} (\underline{E} \times \underline{H}^*)$$

$$P = \int_S \underline{S} \cdot \hat{n} dS = \text{complex power flow through } S$$

Real power = watts = $\text{Re}(P) = \langle \mathcal{P}(t) \rangle$ = average power flow

Reactive power = vars = $\text{Im}(P)$

Transmission Lines in the Time Domain (Lossless)

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)} \quad [\text{F/m}]$$

$$L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \quad [\text{H/m}]$$

$$LC = \mu\epsilon = \frac{1}{c_d^2}$$

$$v(z, t) = F(z - c_d t) + G(z + c_d t)$$

$$i(z, t) = \frac{1}{Z_0} [F(z - c_d t) - G(z + c_d t)]$$

$$v(z, t) = f(t - z / c_d) + g(t + z / c_d)$$

$$i(z, t) = \frac{1}{Z_0} [f(t - z / c_d) - g(t + z / c_d)]$$

$$v^+(z, t) = v_g(t - z / c_d)$$

$$Z_0 = \sqrt{\frac{L}{C}} \quad (\text{lossless line})$$

$$\Gamma_g = \left(\frac{R_g - Z_0}{R_g + Z_0} \right) \quad \Gamma_L = \left(\frac{R_L - Z_0}{R_L + Z_0} \right)$$

$$\Gamma^I = -\Gamma$$

$$T_J = 1 + \Gamma_J$$

$$V^+ = \left(\frac{Z_0}{R_g + Z_0} \right) V_0$$

$$\begin{aligned} v(z, t) = & Av_g(t - z/c_d) \\ & + \Gamma_L Av_g(t - L/c_d - (L - z)/c_d) \\ & + \Gamma_g \Gamma_L Av_g(t - 2L/c_d - z/c_d) \\ & + \Gamma_g \Gamma_L^2 Av_g(t - 3L/c_d - (L - z)/c_d) \\ & + \dots \end{aligned}$$

$$\Gamma_L(t) = 1 - 2e^{-(t-T)/\tau}, \quad t \geq T \quad \tau = Z_0 C_L$$

$$\Gamma_L(t) = -1 + 2e^{-(t-T)/\tau}, \quad t \geq T \quad \tau = L_L / Z_0$$

Transmission Lines in the Frequency Domain

Coax:

$$C = \frac{2\pi\epsilon_0\epsilon_r}{\ln\left(\frac{b}{a}\right)} \quad [\text{F/m}]$$

$$L = \frac{\mu_0}{2\pi} \ln\left(\frac{b}{a}\right) \quad [\text{H/m}]$$

$$G = \frac{2\pi\sigma_d}{\ln\left(\frac{b}{a}\right)} = \frac{2\pi(\omega\epsilon_0\epsilon_r \tan\delta_d)}{\ln\left(\frac{b}{a}\right)} \quad [\text{S/m}]$$

$$R = \left(\frac{1}{2\pi a \sigma_{ma} \delta_{ma}} + \frac{1}{2\pi b \sigma_{mb} \delta_{mb}} \right) \quad [\Omega/\text{m}]$$

$$\delta_m = \sqrt{\frac{2}{\omega\mu_m \sigma_m}} \quad (\mu_m = \mu_0\mu_{rm})$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$\gamma = \alpha + j\beta$$

$$k_z = -j\gamma = \beta - j\alpha$$

$$v_p = \frac{\omega}{\beta}$$

$$\beta = \frac{2\pi}{\lambda_g} \quad (\lambda_g = \lambda_d \text{ for a lossless line})$$

$$\lambda_d = \frac{\lambda_0}{\sqrt{\epsilon_r}} \quad (\text{for } \mu_r = 1)$$

$$\lambda_0 = \frac{c}{f}$$

$$c \equiv 2.99792458 \times 10^8 \text{ [m/s]}$$

$$\text{Attenuation} = \left(\frac{20}{\ln 10} \right) \alpha = (8.6859) \alpha \quad [\text{dB/m}]$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

$$Z_0 = \sqrt{\frac{L}{C}} \quad (\text{lossless line})$$

$$V(z) = A e^{-\gamma z} + B e^{+\gamma z}$$

$$I(z) = \left(\frac{1}{Z_0} \right) [A e^{-\gamma z} - B e^{+\gamma z}]$$

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$\Gamma_L = |\Gamma_L| e^{j\phi}$$

$$V(z) = A (e^{-\gamma z} + \Gamma_L e^{+\gamma z})$$

$$I(z) = \frac{1}{Z_0} A (e^{-\gamma z} - \Gamma_L e^{+\gamma z})$$

$$Z_{\text{in}}(z) = \frac{V(z)}{I(z)}$$

$$Z_{\text{in}} = Z_0 \left(\frac{1 + \Gamma(z)}{1 - \Gamma(z)} \right)$$

$$\Gamma(z) = \Gamma_L e^{j2\beta z} = \Gamma_L e^{-j2\beta d} \quad (\text{lossless line})$$

$$Z_{\text{in}} = Z_0 \left(\frac{1 + \Gamma_L e^{j2\beta z}}{1 - \Gamma_L e^{j2\beta z}} \right) = Z_0 \left(\frac{1 + \Gamma_L e^{-j2\beta d}}{1 - \Gamma_L e^{-j2\beta d}} \right) \quad (\text{lossless line})$$

$$Z_{\text{in}} = Z_0 \left(\frac{Z_L + jZ_0 \tan(\beta d)}{Z_0 + jZ_L \tan(\beta d)} \right) \quad (\text{lossless line})$$

$$Z_{\text{in}} = jZ_0 \tan(\beta l) \quad (\text{short-circuit stub})$$

$$Z_{\text{in}} = -jZ_0 \cot(\beta l) \quad (\text{open-circuit stub})$$

$$|V(z)| = |A| \left| 1 + |\Gamma_L| e^{+j(\phi + 2\beta z)} \right|$$

$$\left| \frac{V(z)}{V^+} \right| = \left| 1 + |\Gamma_L| e^{+j(\phi + 2\beta z)} \right| = |1 + \Gamma(z)|$$

$$\text{VSWR} \equiv \frac{V_{\text{max}}}{V_{\text{min}}}$$

$$\text{VSWR} = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

$$\text{SWR} = \max \left(\frac{R_L}{Z_0}, \frac{Z_0}{R_L} \right) \quad (\text{real load})$$

$$V_{\max} \text{ when } \phi + 2\beta z = 0, \pm 2\pi, \dots$$

$$V_{\min} \text{ when } \phi + 2\beta z = \pm\pi, \pm 3\pi, \dots$$

Impedance Matching

$$Z_{\text{in}} = \frac{Z_{0T}^2}{Z_L}$$

$$Z_{0T} = \sqrt{Z_0 R_L}$$

Smith Charts

$$\Gamma_L = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$Z_{\text{in}}^N(z) = \left(\frac{1 + \Gamma(z)}{1 - \Gamma(z)} \right)$$

$$\Gamma(z) = \Gamma_L e^{+j2\beta z} = \Gamma_L e^{-j2\beta d}$$

$$\Delta\theta = 4\pi \left(\frac{d}{\lambda_d} \right)$$

$$\Gamma'(z) = -\Gamma(z)$$

$$Z_{\text{in}}^N(B) = \frac{1}{Z_{\text{in}}^N(A)}$$

$$\text{SWR} = R_{\text{in}}^N \Big|_{\text{positive real axis}}$$

Plane Waves

Lossless Media

$$\underline{E} = \underline{\hat{x}} E_0 e^{-jkz}$$

$$\underline{H} = \underline{\hat{y}} \frac{1}{\eta} E_0 e^{-jkz}$$

$$\eta = \sqrt{\frac{\mu}{\varepsilon}}$$

$$\eta_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} = 376.7303 \text{ } [\Omega]$$

$$k = \omega \sqrt{\mu \varepsilon}$$

Lossy Media

$$\varepsilon_c \equiv \varepsilon - j \left(\frac{\sigma}{\omega} \right) = \varepsilon'_c - j \varepsilon''_c$$

$$\tan \delta \equiv \frac{\varepsilon''_c}{\varepsilon'_c}$$

$$\tan \delta = \left(\frac{\sigma}{\omega \varepsilon} \right) = \left(\frac{\sigma}{\omega \varepsilon_0 \varepsilon_r} \right)$$

$$\varepsilon_c = \varepsilon_0 \varepsilon_r (1 - j \tan \delta)$$

$$\sigma_d = \omega \varepsilon \tan \delta = \omega \varepsilon_0 \varepsilon_r \tan \delta$$

$$k = \omega \sqrt{\mu \varepsilon_c}$$

$$\eta = \sqrt{\frac{\mu}{\varepsilon_c}}$$

$$k = k' - jk''$$

$$d_p \equiv 1 / k''$$

Good Conductors

$$\delta = d_p = \sqrt{\frac{2}{\omega \mu \sigma}}$$

$$k' \approx k'' \approx \frac{1}{\delta}$$

$$Z_s = \left(\frac{1}{\sigma \delta} \right) (1 + j) = R_s (1 + j) \quad [\Omega]$$

$$R_s = \sqrt{\frac{\omega \mu}{2 \sigma}} \quad [\Omega]$$

$$R = X = R_s \left(\frac{l}{2\pi a} \right)$$

Arbitrary Polarization

$$\underline{E}(z) = (\underline{\hat{x}} E_x + \underline{\hat{y}} E_y) e^{-jkz}$$

$$E_x = a = \text{real number}$$

$$E_y = b e^{j\beta}$$

$$\mathcal{E}_x = \text{Re}(a e^{j\omega t}) = a \cos(\omega t)$$

$$\mathcal{E}_y = \text{Re}(b e^{j\beta} e^{j\omega t}) = b \cos(\omega t + \beta)$$

$$(a) \quad 0 < \beta < \pi \quad \text{LHEP}$$

$$(b) \quad -\pi < \beta < 0 \quad \text{RHEP}$$

Rotation Rule:

In time, the electric field vector rotates from the leading axis to the lagging axis.

Axial ratio and tilt angle

$$\gamma \equiv \tan^{-1} \left(\frac{b}{a} \right)$$

$$0 \leq \gamma \leq 90^\circ$$

$$\sin 2\xi = \sin 2\gamma \sin \beta$$

$$-45^\circ \leq \xi \leq +45^\circ$$

$$\text{AR} = |\cot \xi|$$

$$\xi > 0: \text{ LHEP}$$

$$\xi < 0: \text{ RHEP}$$

$$\tan 2\tau = \tan 2\gamma \cos \beta$$

Reflection

$$\theta_i = \theta_r$$

$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

$$n_1 = \sqrt{\epsilon_{r1} \mu_1} \quad n_2 = \sqrt{\epsilon_{r2} \mu_2}$$

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

$$\alpha_{zt} = k_0 \sqrt{n_1^2 \sin^2 \theta_i - n_2^2}$$

$$\% \text{ power reflected} = 100 |\Gamma|^2$$

$$\Gamma_{\text{TM}} = \frac{Z_2^{\text{TM}} - Z_1^{\text{TM}}}{Z_2^{\text{TM}} + Z_1^{\text{TM}}} \quad \Gamma_{\text{TE}} = \frac{Z_2^{\text{TE}} - Z_1^{\text{TE}}}{Z_2^{\text{TE}} + Z_1^{\text{TE}}}$$

$$Z_1^{\text{TE}} = \left(\frac{\omega \mu_1}{k_{zi}} \right) \quad Z_2^{\text{TE}} = \left(\frac{\omega \mu_2}{k_{zt}} \right)$$

$$Z_1^{\text{TM}} = \left(\frac{k_{zi}}{\omega \epsilon_1} \right) \quad Z_2^{\text{TM}} = \left(\frac{k_{zt}}{\omega \epsilon_2} \right)$$

$$k_{z1} = k_{zi} = \sqrt{k_1^2 - k_x^2} = \sqrt{k_1^2 - (k_1 \sin \theta_i)^2} = k_1 \cos \theta_i$$

$$k_{z2} = k_{zt} = \sqrt{k_2^2 - k_x^2} = \sqrt{k_2^2 - (k_1 \sin \theta_i)^2} = \sqrt{k_2^2 - (k_2 \sin \theta_t)^2} = k_2 \cos \theta_t$$

$$\theta_i = \theta_b = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}}$$

Waveguides

$$\text{TM}_z^{(m,n)} : m > 0, n > 0$$

$$\text{TE}_z^{(m,n)} : m \geq 0, n \geq 0, (m, n) \neq (0, 0)$$

$$k_z = \left(k^2 - \left(\frac{m\pi}{a} \right)^2 - \left(\frac{n\pi}{b} \right)^2 \right)^{1/2}$$

$$k_z = \beta - j\alpha$$

$$f_c = \frac{c_d}{2\pi} \sqrt{\left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2}$$

$$\beta = \sqrt{k^2 - \left(\frac{m\pi}{a} \right)^2 - \left(\frac{n\pi}{b} \right)^2}, \quad f > f_c$$

$$\alpha = \sqrt{\left(\frac{m\pi}{a} \right)^2 + \left(\frac{n\pi}{b} \right)^2 - k^2}, \quad f < f_c$$

$$\text{TE}_{10} : H_z^{(1,0)}(x, y, z) = A_{10} \cos\left(\frac{\pi x}{a}\right) e^{-jk_z z}$$

$$\lambda_g = \frac{2\pi}{\beta}$$

$$\lambda_g = \frac{\lambda_d}{\sqrt{1 - (f_c / f)^2}}$$

$$P_z = \left(\frac{ab}{4\omega\mu} \right) \beta |E_{10}|^2$$

Antennas

$$\underline{E}(r, \theta, \phi) = \left(\frac{e^{-jk_0 r}}{r} \right) \underline{E}^{\text{FF}}(\theta, \phi)$$

$$P_{\text{rad}} = \frac{1}{2\eta_0} \int_0^{2\pi} \int_0^\pi \left| \underline{E}^{\text{FF}}(\theta, \phi) \right|^2 \sin \theta d\theta d\phi$$

$$D(\theta, \phi) \equiv \frac{S_r(\theta, \phi)}{P_{\text{rad}} / (4\pi r^2)} \quad r \rightarrow \infty$$

$$G(\theta, \phi) \equiv e_r D(\theta, \phi)$$

$$D(\pi/2, \phi) = 1.643 \quad (\text{resonant dipole antenna})$$

$$S_r(\theta, \phi) = \left[P_{\text{in}} / (4\pi r^2) \right] G(\theta, \phi)$$

$$P_L = A_{\text{eff}} P_d^{\text{inc}}$$

$$A_{\text{eff}} = G \left(\frac{\lambda_0^2}{4\pi} \right)$$

$$A_{\text{eff}} = A_{\text{phy}} e_{\text{ap}} \quad (\text{dish})$$

$$R_{\text{rad}} \approx 73 \, [\Omega] \quad (\text{resonant dipole antenna})$$

$$Z_{\text{Th}} = Z_{\text{in}}$$